



EDUCATION AND SCHOLASTIC ACHIEVEMENTS			
Program	Institution	CGPA	Year of Completion
M.S (by research) -Thermal	Indian Institute of Technology Madras	8.71/10.0	2027(expected)
B.E Mechanical Engineering	KLE Technological University	9.06/10.0	2023
Pre-University	Impulse PU Science College	91.16%	2019
X (CBSE)	St. Anthony's Public School	9.8/10.0	2017
• Qualified GATE ¹ (ME ²) 2025 among 62000 students.			
TECHNICAL AND SOFTWARE SKILLS			
• Simulation & Analysis: ANSYS Workbench (Structural FEA), ANSYS Fluent, Hypermesh, LAMMPS (Linux-based HPC environment). • CAD & PLM: Siemens NX, SolidWorks (CSWA), Teamcenter. • Semiconductor Manufacturing Processes: Photomask fabrication, wet chemical etching. • Programming Languages: Python, C++			
WORK EXPERIENCE			
Mechanical Engineer Applied Materials India (Jul 2023- Jul 2025)	• Performed structural FEA simulations in ANSYS Workbench to validate mechanical designs of Semiconductor manufacturing equipment. • Conducted CFD and fluid flow analysis in ANSYS Fluent for wet etching processes in photomask fabrication, optimizing chemical distribution, flow uniformity, and process chamber performance to improve etch quality and yield. • Managed Product Lifecycle Management (PLM) releases using Siemens Teamcenter , streamlining engineering change orders (ECOs) and ensuring accurate documentation for cleanroom manufacturing environments. • Authored Operational Method Sheets (OMS) for the assembly of high-precision semiconductor equipment, improving assembly accuracy and reducing rework. • Collaborated with cross-functional teams in design engineering, process engineering, and manufacturing to deliver projects on schedule while meeting stringent cleanroom and process control requirements.		
Intern U.R. Rao Satellite Centre, ISRO (Jan 2023- Jun 2023)	• Structural and Modal analysis of a Satellite Equipment Panel for Static and Dynamic Loads. • Studied the various types of loads acting on the satellite equipment panel during the launch and in-orbit conditions. • Worked on MSC.NASTRAN (Solver), MSC.PATRAN (pre/post processor).		
PROJECTS			
M S Research Project Guide: Dr. Sree Hari P. D. (Jan 2026 – Present)	Self- heating effect (SHE) and thermal transport in Gate All Around (GAA) transistors.		
	• Developed a 3D atomistic model of sub-3 nm GAA transistor in LAMMPS for nanoscale heat transport and self-heating analysis. • Investigating anisotropic thermal conductivity and thermal boundary resistance using NEMD simulations of confined nanosheets and crystalline-amorphous interfaces. • Working on MD–continuum multiscale framework for integrating atomistic thermal parameters into electro-thermal reliability analysis .		
Wet Etching Process Optimization (CIP³) (Dec 2024-June 2025)	• Led process analysis and simulation for wet etching of photoresist on photomask substrates using the piranha reaction (H ₂ SO ₄ + H ₂ O ₂) as part of a customer improvement initiative. • Developed a novel solution to address non-uniform etching, integrating design and process modifications. • Utilized ANSYS Fluent to model chemical distribution and flow patterns during the reaction.		
Prediction of failure of Dual Zone heater (Jul 2026- Sept 2026)	• Performed thermo-structural FEA in ANSYS Workbench to evaluate a dual-zone Low Pressure Chemical Vapor Deposition (LPCVD) ceramic heater operating at temperatures up to 800 °C. • Analysed thermal gradients and mechanical stresses to determine the maximum safe power input without inducing structural failure. • Coordinated with design engineers to align simulation results with operational and material constraints.		
3D-Printed PLA Nanocomposites (June 2022-Dec 2022)	• Performed mechanical testing for 0.1 wt.% graphene, hBN and other nanofiller-reinforced PLA composites. • Demonstrated up to 86.08% higher tensile strength , 90.36% higher tensile modulus , 91.21% higher flexural strength and 69.46% higher fracture toughness over neat PLA. • Established relationships between nanofiller–matrix interactions, fracture morphology and thermal behavior using FTIR, SEM and DSC analysis.		
POSITIONS OF RESPONSIBILITY			
Avishkar Hyperloop	• Designed and Implemented Linear Induction Motor (LIM) Cooling system for the Avishkar hyperloop pod. • Presented full scale research on “Low Pressure Boiling as a Heat exchanger for Hyperloop” in EHW ⁴ 2026. • Achieved 5 th rank overall out of 21 teams participated from worldwide.		
Teaching Assistant (Jan 2026- Nov 2026)	• Conducted computational sessions of Molecular Dynamics Simulations for the course ME5131. • Conducted lab sessions on “Double Pipe Heat Exchanger Experiment” for the course ME3483.		
COURSEWORK			
Advanced Heat and Mass Transfer	Incompressible Fluid Flow	Nanoscale Thermo-fluidics	
Applications of Molecular Dynamics*	Fundamentals Of CFD*	Finite Element Analysis	
CERTIFICATIONS			
SOLIDWORKS CSWA	GD&T training - CMTI	Pre-Processing with HyperWorks	
PUBLICATIONS AND PATENTS			
PUBLICATION	• “Additive Manufacturing of PLA Composites: Impact of Nanofillers on Mechanical, Fracture and Thermal Behavior” - Materials Research Express, May-2026. • Contribution-Mechanical Testing, SEM, DSC, FTIR data interpretation.		
PATENT (IDF 3723)	• Mechanical Radial Constriction External Iris Brake for Automotive Vehicles (filed, February 2026).		

1-Graduate Aptitude Test in Engineering, 2-Mechanical Engineering,3- Customer Improvement Project ,4-European Hyperloop Week, *-Ongoing.